LLM COURSE PROJECT · PROPOSAL

ScamLoop-HE

Attacker-defender training for robust Hebrew SMS scam detection

Attacker LLM vs Defender classifier



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1 · MOTIVATING USE CASE

Hebrew scam SMS keep evolving

Background

Israelis receive a steady stream of SMS impersonating the post, banks, tax authority and government services.

Why it matters

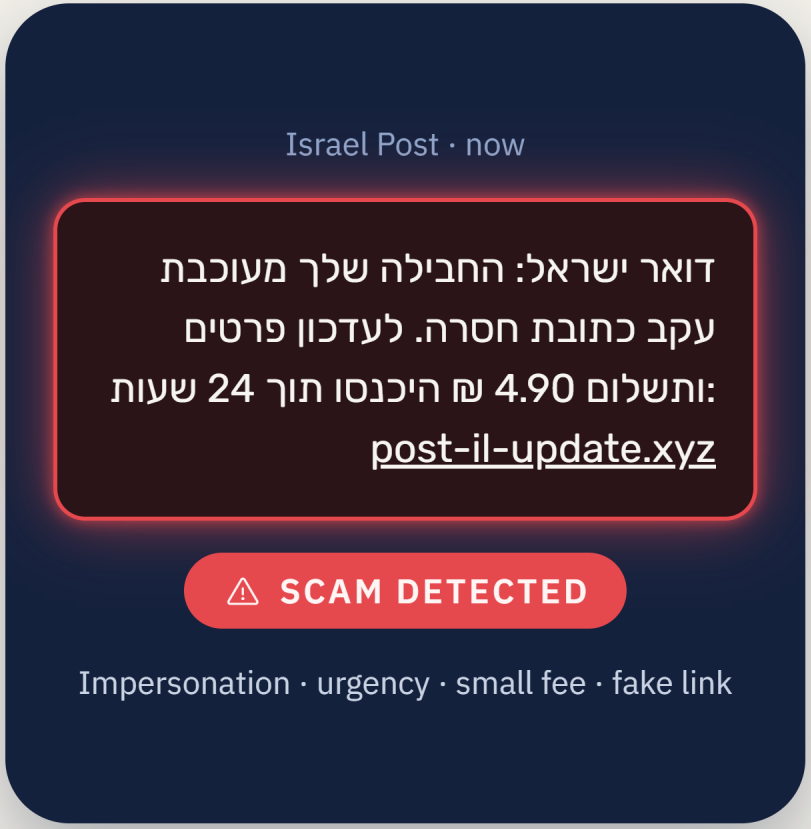
One click can lead to stolen card details and accounts, and a convincing message can fool anyone.

Why it’s hard

Little labeled Hebrew data, and scammers keep rewording to slip past filters.

Today

Generic spam filters and public warnings. We found no Hebrew scam dataset to train on.



+ RELATED WORK AND GAP

What exists, and what is missing

Work	What they did	Language	Gap
Scam Shield (Chang et al., 2025)	LLM rewrites scams to look legit; one fine-tuning pass	English	No iterative retraining
Adversarial Robustness in Smishing Detection (arXiv 2608.12889)	Evasive SMS made with Gemini to attack classifiers	English, Swahili	Retraining left as future work
COVA (arXiv 2604.11752)	Attacker and victim LLM agents generate scam dialogues	English	No detector in the loop
SMS Spam Multilingual Collection	Spam SMS machine-translated to 21 languages	No Hebrew	Translated, no Hebrew

Gap: no labeled Hebrew smishing data, and no attacker-defender loop that retrains the detector each round.

Searched Nov 2026: Hugging Face, Kaggle, arXiv, Hebrew NLP resource lists (NNLP-IL, NLPH)

2 · PROJECT TASK

Classify a Hebrew SMS - and stay robust

INPUT

One Hebrew SMS message (text, links, numbers)

OUTPUT

Scam / legitimate (main) + scam type: delivery, bank, government, prize, other (secondary)

RESEARCH QUESTION

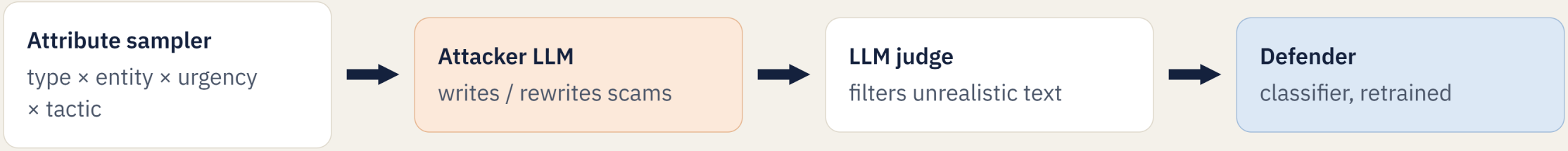
Does iterative adversarial training on synthetic Hebrew SMS data improve smishing detection on independently collected real Hebrew messages, compared with native-Hebrew training alone and a translated-English baseline?

Novelty 1. No public labeled Hebrew scam SMS set found; multilingual spam sets are machine-translated and skip Hebrew.

Novelty 2. Iterative loop that retrains the defender; prior work only attacks or trains once.

Secondary. Does it also catch a scam type held out from training?

An attacker-defender loop



Each round the attacker rewrites 200 caught scams. A message **evades** if the defender labels it legitimate. The defender retrains on the evading ones plus a **replay buffer** of earlier rounds.

Role	Models	Adjustment
Baselines	DictaBERT trained on translated English data; GPT few-shot for reference	Translation / prompting
Fine-tuned defenders	DictaBERT (main, Hebrew BERT); SetFit only if time allows	Full fine-tune; few-shot contrastive
Attacker	Open LLM run locally (Ollama) + GPT via API	Prompted, feedback from defender

4 · DATA SPECIFICATION AND GENERATION

Label first, then generate

STEP 1 · Pick label + attributes

Label: scam
Type: delivery
Entity: Israel Post
Urgency: 24 hours
Tactic: fake short link

STEP 2 · LLM writes the SMS

דואר ישראל: החבילה שלך מעוכבת. לעדכון
כתובת תוך 24 שעות: bit.ly/il-post7

STEP 3 · Filter

LLM judge drops unrealistic messages; we review a sample by hand.

The label is chosen **before** the text, so every message is labeled automatically; the hand-checked sample also confirms the label fits.

Synthetic scams
2,000 base + up to 1,000 from rounds (5 × 200)

Legitimate look-alikes
2,000 bank / HMO-style notices; legit by verified sender, not by domain

Real test set
Target: 150 real (75 / 75), to be collected

+ EXPERIMENTAL DESIGN

Same defender, same test set, four conditions

1 · **Translated English** (baseline)

2 · **Native Hebrew** synthetic only

3 · **Native + static attacker data** made in advance

4 · **Native + attacker loop** (ours)

Defender: DictaBERT in all four. Conditions 3 and 4 get the **same amount** of attacker data, so the only difference is feedback.

REAL TEST SET PLAN

Sources: scam SMS we and our families received; public warning posts; legit notices from verified senders.

Labeling: both of us label separately, resolve disagreements, report Cohen’s kappa.

Privacy: remove names, numbers and personal links.

Fallback: if under 150, report as a pilot with confidence intervals.

Leakage control: train/val/test split by generation template; near-duplicates removed; the real test set is never used for training or model selection.

5 · METRICS AND KPIS

How we will know it works

Does it catch scams?	**Precision, recall, F1** on the real test set (main); on a held-out scam type (secondary).
Does it survive new tricks?	**Evasion rate**: % of the 200 new attacker scams per round labeled legit. Should drop; stop below 5% or after 5 rounds.
Is our fake data realistic?	**LLM judge** and a human sample rate realism; Cohen's kappa measures judge-human agreement.
Why did it decide that?	**SHAP** highlights the words behind each decision, especially the mistakes.

MAIN COMPARISON: Translated → Native → Native + static → **Native + loop**